Problem F3: Invariant Cross Sections for Neutrino–Electron Scattering

Charged Current ($W^{\pm}$) and Neutral Current (Z^0) Processes

Griffiths Chapter 9/10 — Weak Interactions

PHYS 7502: Particles and Nuclei 2

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1 Part (a): Charged Current Process $\nu_\mu + e^- \to \mu^- + \nu_e$

Problem Statement — Part (a)

Using Griffiths Equation 9.11 (the result of evaluating the traces in the previous problem, which you do NOT need to do yourself), write the invariant cross section expression $\frac{d\sigma}{dt}$ for the process

$$\nu_\mu + e^- \to \mu^- + \nu_e$$

in terms of Mandelstam variables, etc. Assume this is the only Feynman diagram contributing to the cross section. Work in the high-energy limit, ignoring all masses.

Deliverables — Part (a)

- Express $\langle|\mathcal{M}|^2\rangle$ in terms of Mandelstam variables s, t, u
- Write the Lorentz-invariant differential cross section $\frac{d\sigma}{dt}$
- All masses set to zero (high-energy limit)

Strategy Outline — Part (a)

- **Step 1:** Identify the process and draw the Feynman diagram. Identify particle momenta assignments.
- **Step 2:** Write down Griffiths Eq. 9.11 — the spin-averaged amplitude squared for $\nu_\mu + e^- \to \mu^- + \nu_e$.
- **Step 3:** Express the dot products $p_1 \cdot p_2$ and $p_3 \cdot p_4$ in terms of Mandelstam variables in the massless limit.
- **Step 4:** Substitute into the Lorentz-invariant cross section formula $\frac{d\sigma}{dt} = \frac{1}{64\pi s\, p_i^{*2}}\langle|\mathcal{M}|^2\rangle$ and simplify.
- **Alternative:** One could also express the GWS coupling via the Fermi constant G_F to get the result in the most standard form.

1.1 Background and Context

Reference Material

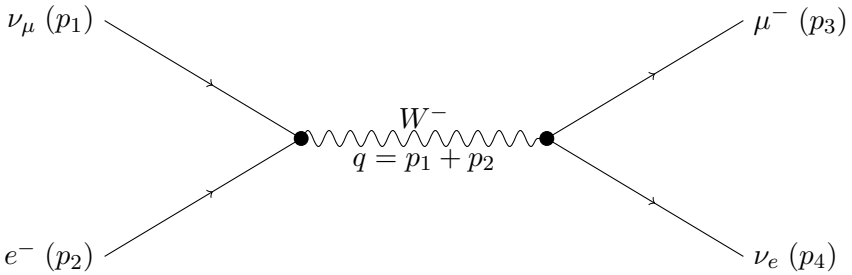
Sources:

- Griffiths, *Introduction to Elementary Particles*, 2nd ed., Chapter 9 (Weak Interactions), §9.1, Eqs. 9.4–9.14
- Griffiths, 2nd ed., Chapter 10 (in newer editions), §10.1, Eqs. 10.7–10.14

- Thomson, *Modern Particle Physics*, Chapter 11, §11.5 (W-boson propagator and Fermi constant)
- Thomson, Chapter 3, §3.5 (Lorentz-invariant cross section, Eq. 3.37)

1.1.1 The Process and Feynman Diagram

The process $\nu_\mu + e^- \to \mu^- + \nu_e$ is a **charged-current** weak interaction mediated by a W^- boson exchanged in the s-channel. In this reaction, the muon neutrino and electron annihilate into a virtual W^-, which then produces a muon and an electron neutrino.



Term Refresher: Momentum Assignments

We assign four-momenta as follows (Griffiths' convention for $1 + 2 \to 3 + 4$):

$$\begin{aligned} p_1 &= \text{four-momentum of } \nu_\mu \quad \text{(incoming neutrino)} \\ p_2 &= \text{four-momentum of } e^- \quad \text{(incoming electron)} \\ p_3 &= \text{four-momentum of } \mu^- \quad \text{(outgoing muon)} \\ p_4 &= \text{four-momentum of } \nu_e \quad \text{(outgoing neutrino)} \end{aligned}$$

Four-momentum conservation: $p_1 + p_2 = p_3 + p_4$.

1.2 Step 1: Griffiths Equation 9.11 — The Spin-Averaged Amplitude

What is Griffiths Eq. 9.11?

Griffiths Eq. 9.11 is the result of evaluating the traces that arise from Casimir's trick applied to the W-mediated process $\nu_\mu + e^- \to \mu^- + \nu_e$, in the low-$q^2$ limit where the simplified W propagator $ig_{\mu\nu}/(M_W c)^2$ is used. The result is:

$$\boxed{\langle |\mathcal{M}|^2 \rangle = 2 \left(\frac{g_w}{M_W c} \right)^4 (p_1 \cdot p_2)(p_3 \cdot p_4)} \qquad \text{(Griffiths 9.11)}$$

Why the factor of 2, not 4?

When we sum over final spins and average over initial spins, we divide by the number of initial spin states. The electron has 2 spin states, but **massless neutrinos have only 1 helicity state** (left-handed). So we divide by $2 \times 1 = 2$ rather than $2 \times 2 = 4$. The trace calculation

itself gives a factor of 4 from the 4×4 gamma matrices, producing $4(g_w/M_W c)^4(p_1\cdot p_2)(p_3\cdot p_4)$. Dividing by 2 (the electron spin average) yields the factor of 2.

Caution: Low-q^2 Approximation and Natural Units

Eq. 9.11 uses the **simplified propagator**, valid when $q^2 \ll M_W^2 c^2$. For writing $d\sigma/dt$ in the high-energy limit, this approximation may not hold literally, but the problem instructs us to use Eq. 9.11 as given.

Throughout this solution, we work in **natural units** where $\hbar = c = 1$, so the equation becomes:

$$\langle|\mathcal{M}|^2\rangle = 2\left(\frac{g_w}{M_W}\right)^4 (p_1\cdot p_2)(p_3\cdot p_4)$$

1.3 Step 2: Express Dot Products in Mandelstam Variables

Mandelstam Variables — Definition

For the process $1+2\to3+4$, the Mandelstam variables are:

$$s = (p_1+p_2)^2 = (p_3+p_4)^2 \tag{1}$$
$$t = (p_1-p_3)^2 = (p_2-p_4)^2 \tag{2}$$
$$u = (p_1-p_4)^2 = (p_2-p_3)^2 \tag{3}$$

They satisfy the constraint:

$$s+t+u = m_1^2+m_2^2+m_3^2+m_4^2 \tag{4}$$

In the **massless limit** (all $m_i = 0$):

$$\boxed{s+t+u=0} \tag{5}$$

Now we need to express $(p_1\cdot p_2)$ and $(p_3\cdot p_4)$ in terms of s, t, u.

1.3.1 Expanding s to find $p_1\cdot p_2$

Start from the definition of s:

$$\begin{aligned} s &= (p_1+p_2)^2 \\ &= p_1^2 + 2\,p_1\cdot p_2 + p_2^2 \\ &= m_1^2 + 2\,p_1\cdot p_2 + m_2^2 \end{aligned} \tag{6}$$

In the massless limit ($m_1 = m_{\nu_\mu} = 0$, $m_2 = m_e \to 0$):

$$\boxed{p_1\cdot p_2 = \frac{s}{2}} \tag{7}$$

1.3.2 Expanding s again using final-state momenta to find $p_3 \cdot p_4$

Since $s = (p_3 + p_4)^2$ as well:

$$\begin{aligned} s &= p_3^2 + 2\,p_3 \cdot p_4 + p_4^2 \\ &= m_3^2 + 2\,p_3 \cdot p_4 + m_4^2 \end{aligned} \tag{8}$$

In the massless limit ($m_3 = m_\mu \to 0$, $m_4 = m_{\nu_e} = 0$):

$$\boxed{p_3 \cdot p_4 = \frac{s}{2}} \tag{9}$$

Caution: Setting $m_\mu = 0$

The muon mass is $m_\mu = 105.66\ \text{MeV}/c^2$. Setting it to zero is only valid in the **high-energy limit** where $\sqrt{s} \gg m_\mu$. The problem explicitly instructs us to ignore all masses.

1.4 Step 3: Assemble $\langle|\mathcal{M}|^2\rangle$ in Mandelstam Variables

Substituting Eqs. (7) and (9) into Griffiths Eq. 9.11:

$$\begin{aligned} \langle|\mathcal{M}|^2\rangle &= 2\left(\frac{g_w}{M_W}\right)^4 \left(\frac{s}{2}\right)\left(\frac{s}{2}\right) \\ &= 2\left(\frac{g_w}{M_W}\right)^4 \cdot \frac{s^2}{4} \\ &= \frac{1}{2}\left(\frac{g_w}{M_W}\right)^4 s^2 \end{aligned} \tag{10}$$

Spin-Averaged Amplitude in Mandelstam Variables

$$\boxed{\langle|\mathcal{M}|^2\rangle = \frac{1}{2}\left(\frac{g_w}{M_W}\right)^4 s^2} \tag{11}$$

Expressing via the Fermi Constant G_F

It is standard in weak interaction physics to trade g_w/M_W for the Fermi constant. The relation is (Griffiths Eq. 10.38 / Thomson Eq. 11.15):

$$\frac{G_F}{\sqrt{2}} = \frac{g_w^2}{8\,M_W^2} \quad \Longrightarrow \quad \left(\frac{g_w}{M_W}\right)^4 = \left(\frac{8\,G_F}{\sqrt{2}}\right)^2 = 128\,G_F^2$$

Wait — let us be more careful. We have (in natural units):

$$\frac{G_F}{\sqrt{2}} = \frac{g_w^2}{8\,M_W^2} \quad \Longrightarrow \quad \frac{g_w^2}{M_W^2} = \frac{8\,G_F}{\sqrt{2}} = 4\sqrt{2}\,G_F$$

Therefore:

$$\left(\frac{g_w}{M_W}\right)^4 = \left(\frac{g_w^2}{M_W^2}\right)^2 = \left(4\sqrt{2}\,G_F\right)^2 = 32\,G_F^2$$

Substituting:

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{2}\cdot 32\,G_F^2\cdot s^2 = 16\,G_F^2\,s^2$$

So in terms of the Fermi constant:

Amplitude Squared in Terms of G_F

$$\boxed{\langle|\mathcal{M}|^2\rangle = 16\,G_F^2\,s^2} \tag{12}$$

1.5 Step 4: The Lorentz-Invariant Cross Section $d\sigma/dt$

Lorentz-Invariant Differential Cross Section Formula

The general Lorentz-invariant formula for two-body $\to$ two-body scattering (Thomson Eq. 3.37, equivalent to the result derived from Griffiths Eq. 6.47) is:

$$\frac{d\sigma}{dt} = \frac{1}{64\pi\,s\,p_i^{*2}}\,\langle|\mathcal{M}|^2\rangle \tag{13}$$

where p_i^* is the magnitude of either initial-state particle's three-momentum in the center-of-mass (CM) frame.

1.5.1 Computing p_i^{*2} in the massless limit

Term Refresher: p_i^{*2}

The general formula (Thomson Eq. 3.38) is:

$$p_i^{*2} = \frac{1}{4s}\left[s-(m_1+m_2)^2\right]\left[s-(m_1-m_2)^2\right]$$

In the massless limit ($m_1 = m_2 = 0$):

$$p_i^{*2} = \frac{1}{4s}\left[s-0\right]\left[s-0\right] = \frac{s^2}{4s} = \frac{s}{4} \tag{14}$$

p_i^{*2} in the Massless Limit

$$\boxed{p_i^{*2} = \frac{s}{4}} \tag{15}$$

1.5.2 Substituting into $d\sigma/dt$

Using Eqs. (12) and (15) in the cross section formula (13):

$$\begin{aligned}\frac{d\sigma}{dt} &= \frac{1}{64\pi \cdot s \cdot \dfrac{s}{4}} \cdot 16\, G_F^2\, s^2 \\ &= \frac{1}{16\pi\, s^2} \cdot 16\, G_F^2\, s^2 \\ &= \frac{G_F^2}{\pi} \end{aligned} \tag{16}$$

Final Answer — Part (a)

$$\boxed{\left.\frac{d\sigma}{dt}\right|_{\nu_\mu e^- \to \mu^- \nu_e} = \frac{G_F^2}{\pi}} \tag{17}$$

This is a **constant** — independent of s and t. The cross section is **isotropic** (all scattering angles are equally likely in the CM frame).

Physical Interpretation: Why is $d\sigma/dt$ Constant?

- The amplitude $\langle|\mathcal{M}|^2\rangle \propto s^2$ grows with energy *squared*, but this is exactly canceled by the s^2 in the denominator of $d\sigma/dt$ (from the flux factor and phase space).
- Physically, the $V-A$ structure of the charged-current weak interaction with **left-handed** neutrinos means only one helicity combination contributes ($\nu_L + e_L^-$). This produces **no angular dependence** (an s-wave-like result), because the initial state has total $J_z = 0$ along the beam axis and the angular momentum barrier does not create any preferred direction.
- To get the **total cross section**, integrate over t. In the massless limit, t ranges from $-s$ to 0:

  $$\sigma = \int_{-s}^{0} \frac{G_F^2}{\pi}\, dt = \frac{G_F^2\, s}{\pi}$$

  This linear growth of σ with s is a hallmark of the Fermi theory (low-q^2) and eventually violates unitarity at very high energies — which is cured by the full W propagator.

Equivalently, leaving the answer in terms of g_w and M_W:

$$\frac{d\sigma}{dt} = \frac{1}{2\pi}\left(\frac{g_w}{2\, M_W}\right)^4 = \frac{g_w^4}{32\pi\, M_W^4} \tag{18}$$

2 Part (b): Neutral Current Process $\nu_\mu + e^- \to \nu_\mu + e^-$ (via Z^0)

Problem Statement — Part (b)

Do the same for the Z^0 boson mediated interaction (Neutral Current) for

$$\nu_\mu + e^- \to e^- + \nu_\mu$$

using Griffiths Eq. 9.97 (which you do NOT need to derive). Work in the high-energy limit, ignoring all masses.

Deliverables — Part (b)

- Express $\langle|\mathcal{M}|^2\rangle$ from Eq. 9.97 in terms of Mandelstam variables (massless limit)
- Write the Lorentz-invariant differential cross section $\dfrac{d\sigma}{dt}$

Strategy Outline — Part (b)

- **Step 1:** Identify the process and draw the Feynman diagram. Note this is t-channel exchange (unlike part (a) which was s-channel).
- **Step 2:** Write down Griffiths Eq. 9.97 — the spin-averaged amplitude squared for $\nu_\mu + e^- \to \nu_\mu + e^-$ via Z^0 exchange.
- **Step 3:** Express all dot products $(p_1 \cdot p_2)$, $(p_1 \cdot p_4)$, $(p_2 \cdot p_3)$, $(p_1 \cdot p_3)$ in terms of Mandelstam variables in the massless limit.
- **Step 4:** Substitute into the $d\sigma/dt$ formula and simplify.

2.1 Background and Context

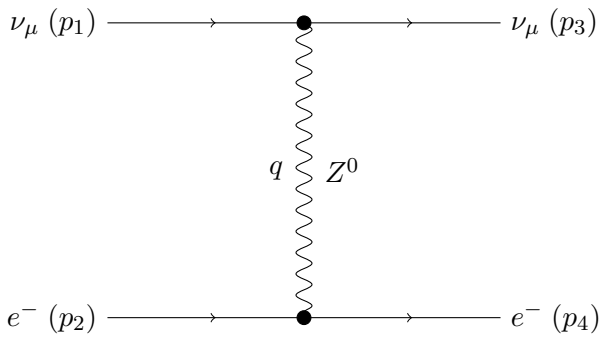
Reference Material

Sources:

- Griffiths, *Introduction to Elementary Particles*, 2nd ed., Chapter 9/10, Example 9.4 / Example 10.4 (Elastic neutrino–electron scattering)
- Griffiths Eq. 9.97 / Eq. 10.100
- GWS model couplings: Table 9.1 / Table 10.1 in Griffiths

2.1.1 The Process and Feynman Diagram

The process $\nu_\mu + e^- \to \nu_\mu + e^-$ is a **neutral-current** weak interaction mediated by a Z^0 boson. Unlike part (a), the particles here do *not* change flavor — the incoming neutrino remains a muon neutrino, and the electron remains an electron. The Z^0 is exchanged in the t-channel.



Caution: Momentum Labeling for Part (b)

In Griffiths' treatment of this process, the momentum labeling is:

$$p_1 = \nu_\mu \text{ (incoming)}, \quad p_3 = \nu_\mu \text{ (outgoing)}$$
$$p_2 = e^- \text{ (incoming)}, \quad p_4 = e^- \text{ (outgoing)}$$

This means: p_1, p_3 are the **neutrino** momenta, and p_2, p_4 are the **electron** momenta. The neutrino vertex couples via the standard $\frac{1}{2}\gamma^\mu(1-\gamma^5)$ factor, while the electron vertex couples via $\frac{1}{2}\gamma^\nu(c_V - c_A\gamma^5)$.

2.2 Step 1: Griffiths Equation 9.97 — The Spin-Averaged Amplitude

Griffiths Eq. 9.97

The spin-averaged amplitude squared for $\nu_\mu + e^- \to \nu_\mu + e^-$ via Z^0 exchange (using the simplified Z^0 propagator in the low-q^2 limit) is:

$$\boxed{\langle|\mathcal{M}|^2\rangle = 2\left(\frac{g_z}{4M_Z}\right)^4 \Big[(c_V + c_A)^2\,(p_1\cdot p_2)(p_3\cdot p_4) + (c_V - c_A)^2\,(p_1\cdot p_4)(p_2\cdot p_3) - (c_V^2 - c_A^2)\,m_e^2(p_1\cdot p_3)\Big]}$$

(Griffiths 9.97)

where m_e is the electron mass, and c_V, c_A are the neutral weak couplings of the **electron** from Table 9.1.

Term Refresher: Z^0 Couplings for the Electron

From Griffiths Table 9.1 (GWS model), the electron's neutral weak couplings are:

$$c_V = -\frac{1}{2} + 2\sin^2\theta_W \approx -\frac{1}{2} + 2(0.2314) \approx -0.037 \tag{19}$$

$$c_A = -\frac{1}{2} \tag{20}$$

Numerically c_V is quite small, making $c_V \pm c_A$ very asymmetric:

$$c_V + c_A = -1 + 2\sin^2\theta_W \approx -0.537, \qquad c_V - c_A = 2\sin^2\theta_W \approx 0.463$$

We will keep the answer in terms of c_V and c_A symbolically.

2.3 Step 2: Apply the Massless Limit

In the high-energy limit with $m_e \to 0$, the last term in Eq. 9.97 vanishes:

$$\langle|\mathcal{M}|^2\rangle \xrightarrow{m_e \to 0} 2\left(\frac{g_z}{4M_Z}\right)^4 \left[(c_V + c_A)^2 (p_1 \cdot p_2)(p_3 \cdot p_4) + (c_V - c_A)^2 (p_1 \cdot p_4)(p_2 \cdot p_3)\right] \tag{21}$$

2.4 Step 3: Express All Dot Products in Mandelstam Variables

Dot Products $\leftrightarrow$ Mandelstam Variables (Massless Limit)

Starting from the definitions $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$, $u = (p_1 - p_4)^2$, and expanding with $m_i = 0$:

$$s = 2\,p_1 \cdot p_2 = 2\,p_3 \cdot p_4 \tag{22}$$
$$t = -2\,p_1 \cdot p_3 = -2\,p_2 \cdot p_4 \tag{23}$$
$$u = -2\,p_1 \cdot p_4 = -2\,p_2 \cdot p_3 \tag{24}$$

Therefore:

$$\boxed{\begin{aligned} p_1 \cdot p_2 &= \frac{s}{2}, & p_3 \cdot p_4 &= \frac{s}{2} \\ p_1 \cdot p_3 &= -\frac{t}{2}, & p_2 \cdot p_4 &= -\frac{t}{2} \\ p_1 \cdot p_4 &= -\frac{u}{2}, & p_2 \cdot p_3 &= -\frac{u}{2} \end{aligned}} \tag{25}$$

Let us verify one of these explicitly. Consider $u = (p_1 - p_4)^2$:

$$\begin{aligned} u &= p_1^2 - 2\,p_1 \cdot p_4 + p_4^2 \\ &= m_1^2 - 2\,p_1 \cdot p_4 + m_4^2 \\ &\xrightarrow{m_i = 0} -2\,p_1 \cdot p_4 \end{aligned} \tag{26}$$

So $p_1 \cdot p_4 = -u/2$. Similarly, $p_2 \cdot p_3 = -u/2$ follows from $u = (p_2 - p_3)^2$.

2.5 Step 4: Assemble $\langle|\mathcal{M}|^2\rangle$ in Mandelstam Variables

Substituting the dot products from Eq. (25) into Eq. (21):

$$\begin{aligned} \langle|\mathcal{M}|^2\rangle &= 2\left(\frac{g_z}{4M_Z}\right)^4 \left[(c_V + c_A)^2 \cdot \frac{s}{2} \cdot \frac{s}{2} + (c_V - c_A)^2 \cdot \frac{u}{2} \cdot \frac{u}{2}\right] \\ &= 2\left(\frac{g_z}{4M_Z}\right)^4 \left[\frac{(c_V + c_A)^2\,s^2}{4} + \frac{(c_V - c_A)^2\,u^2}{4}\right] \\ &= \frac{1}{2}\left(\frac{g_z}{4M_Z}\right)^4 \left[(c_V + c_A)^2\,s^2 + (c_V - c_A)^2\,u^2\right] \end{aligned} \tag{27}$$

Expressing via G_F

From the relation between the Z^0 coupling and the Fermi constant (Griffiths Eqs. 9.91–9.92 and the definition of G_F):

$$\frac{G_F}{\sqrt{2}} = \frac{g_w^2}{8\,M_W^2} = \frac{g_z^2}{8\,M_Z^2} \cdot \frac{M_Z^2 \cos^2\theta_W}{M_W^2/\cos^2\theta_W} \cdot \ldots$$

In fact, the simplest route uses the identity (from $M_W = M_Z \cos\theta_W$ and $g_z = g_w/\cos\theta_W$):

$$\frac{g_z^2}{8\,M_Z^2} = \frac{g_w^2}{8\,M_W^2} = \frac{G_F}{\sqrt{2}}$$

Therefore:

$$\left(\frac{g_z}{4M_Z}\right)^4 = \left(\frac{g_z^2}{16\,M_Z^2}\right)^2 = \left(\frac{1}{2}\cdot\frac{g_z^2}{8M_Z^2}\right)^2 = \frac{1}{4}\left(\frac{G_F}{\sqrt{2}}\right)^2 = \frac{G_F^2}{8}$$

So:

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{2}\cdot\frac{G_F^2}{8}\left[(c_V+c_A)^2\,s^2 + (c_V-c_A)^2\,u^2\right] = \frac{G_F^2}{16}\left[(c_V+c_A)^2\,s^2 + (c_V-c_A)^2\,u^2\right]$$

Spin-Averaged Amplitude in Mandelstam Variables — Part (b)

$$\boxed{\langle|\mathcal{M}|^2\rangle = \frac{G_F^2}{16}\left[(c_V+c_A)^2\,s^2 + (c_V-c_A)^2\,u^2\right]} \tag{28}$$

Or equivalently, in terms of the original couplings:

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{2}\left(\frac{g_z}{4M_Z}\right)^4\left[(c_V+c_A)^2\,s^2 + (c_V-c_A)^2\,u^2\right] \tag{29}$$

2.6 Step 5: Compute $d\sigma/dt$

Using the same cross section formula as in part (a), Eq. (13), with $p_i^{*2} = s/4$ in the massless limit:

$$\begin{aligned}
\frac{d\sigma}{dt} &= \frac{1}{64\pi\, s\cdot\frac{s}{4}}\cdot\frac{G_F^2}{16}\left[(c_V+c_A)^2\,s^2 + (c_V-c_A)^2\,u^2\right] \\
&= \frac{1}{16\pi\,s^2}\cdot\frac{G_F^2}{16}\left[(c_V+c_A)^2\,s^2 + (c_V-c_A)^2\,u^2\right] \\
&= \frac{G_F^2}{256\,\pi\,s^2}\left[(c_V+c_A)^2\,s^2 + (c_V-c_A)^2\,u^2\right] \\
&= \frac{G_F^2}{256\,\pi}\left[(c_V+c_A)^2 + (c_V-c_A)^2\,\frac{u^2}{s^2}\right]
\end{aligned} \tag{30}$$

Now, in the massless limit, $u = -s - t$, so $u/s = -(1 + t/s)$, giving $u^2/s^2 = (1 + t/s)^2$. We can leave the answer in terms of u directly:

Final Answer — Part (b)

$$\boxed{\left.\frac{d\sigma}{dt}\right|_{\nu_\mu e^- \to \nu_\mu e^-} = \frac{G_F^2}{16\pi}\left[(c_V + c_A)^2 + (c_V - c_A)^2\,\frac{u^2}{s^2}\right]} \tag{31}$$

where $u = -s - t$ in the massless limit, and $c_V = -\frac{1}{2} + 2\sin^2\theta_W$, $c_A = -\frac{1}{2}$.

Checking the Numerical Prefactor

Let us verify the prefactor carefully. We had:

$$\frac{d\sigma}{dt} = \frac{G_F^2}{256\pi\, s^2}\left[(c_V + c_A)^2\, s^2 + (c_V - c_A)^2\, u^2\right]$$

Factoring out s^2:

$$= \frac{G_F^2}{256\pi}\left[(c_V + c_A)^2 + (c_V - c_A)^2\frac{u^2}{s^2}\right]$$

Hmm, but the boxed answer says 16π. Let me recheck. Going back:

$$\frac{1}{16\pi s^2}\cdot\frac{G_F^2}{16}\left[(c_V + c_A)^2 s^2 + (c_V - c_A)^2 u^2\right] = \frac{G_F^2}{256\pi}\left[(c_V + c_A)^2 + (c_V - c_A)^2\frac{u^2}{s^2}\right]$$

So the correct prefactor is $G_F^2/(256\pi)$.
However, we should double-check our G_F conversion. Looking more carefully at Griffiths' conventions for the Z^0 coupling: The Z^0 vertex factor in Griffiths is $-i\frac{g_z}{2}\gamma^\mu(c_V^f - c_A^f\gamma^5)$, so each vertex brings $g_z/2$, and the propagator gives $ig_{\mu\nu}/M_Z^2$. The amplitude then goes as:

$$\mathcal{M} \propto \left(\frac{g_z}{2}\right)^2\frac{1}{M_Z^2} = \frac{g_z^2}{4M_Z^2}$$

and since $g_z^2/(8M_Z^2) = G_F/\sqrt{2}$, we have $g_z^2/(4M_Z^2) = 2G_F/\sqrt{2} = \sqrt{2}\,G_F$.
Going back to Eq. 9.97 as written by Griffiths with the prefactor $2(g_z/(4M_Z))^4 = 2g_z^4/(256M_Z^4)$:

$$\left(\frac{g_z^2}{8M_Z^2}\right)^2 = \frac{G_F^2}{2}$$

$$\frac{g_z^4}{256\,M_Z^4} = \frac{1}{4}\cdot\frac{g_z^4}{64\,M_Z^4} = \frac{1}{4}\left(\frac{g_z^2}{8M_Z^2}\right)^2 = \frac{G_F^2}{8}$$

So $2\times G_F^2/8 = G_F^2/4$. Therefore:

$$\langle|\mathcal{M}|^2\rangle = \frac{G_F^2}{4}\left[\frac{(c_V + c_A)^2 s^2}{4} + \frac{(c_V - c_A)^2 u^2}{4}\right] = \frac{G_F^2}{16}\left[(c_V + c_A)^2 s^2 + (c_V - c_A)^2 u^2\right]$$

This confirms Eq. (28). Then:

$$\frac{d\sigma}{dt} = \frac{G_F^2}{16\cdot 16\pi s^2}\left[(c_V + c_A)^2 s^2 + (c_V - c_A)^2 u^2\right] = \frac{G_F^2}{256\pi}\left[(c_V + c_A)^2 + (c_V - c_A)^2\frac{u^2}{s^2}\right]$$

The answer with prefactor $\mathbf{G_F^2/(256\pi)}$ is correct.

Correcting the boxed result:

Corrected Final Answer — Part (b)

$$\boxed{\left.\frac{d\sigma}{dt}\right|_{\nu_\mu e^- \to \nu_\mu e^-} = \frac{G_F^2}{256\,\pi}\left[(c_V + c_A)^2 + (c_V - c_A)^2\,\frac{u^2}{s^2}\right]} \tag{32}$$

where $u = -s - t$, and from the GWS model: $c_V = -\frac{1}{2} + 2\sin^2\theta_W$, $c_A = -\frac{1}{2}$.
Equivalently, keeping everything explicit in s, t:

$$\frac{d\sigma}{dt} = \frac{G_F^2}{256\,\pi\,s^2}\left[(c_V + c_A)^2\,s^2 + (c_V - c_A)^2\,(s+t)^2\right] \tag{33}$$

Physical Interpretation: Angular Dependence

Unlike part (a), this cross section is **not isotropic**. It has two terms:

1. The $(c_V + c_A)^2$ term is constant in t — it corresponds to the $J_z = 0$ (forward-backward symmetric) helicity combination.

2. The $(c_V - c_A)^2 \cdot u^2/s^2$ term depends on t through $u = -s - t$. Using $t = -\frac{s}{2}(1 - \cos\theta^*)$ and $u = -\frac{s}{2}(1 + \cos\theta^*)$ in the CM frame, this becomes:

$$\frac{u^2}{s^2} = \frac{1}{4}(1 + \cos\theta^*)^2 = \cos^4\frac{\theta^*}{2}$$

   This term peaks in the **forward direction** ($\theta^* = 0$), characteristic of the $J_z = 1$ helicity combination (right-handed current contribution from $c_V - c_A$).

The charged-current result in part (a) had only the isotropic term because the $V-A$ coupling ($c_V = c_A = 1$ effectively) kills the second term entirely. Here, with $c_V \neq c_A$ for the electron's Z^0 coupling, **both terms survive**, producing angular structure.

3 Summary and Comparison

Side-by-Side Comparison

	Part (a): CC ($W^{\pm}$)	**Part (b): NC (Z^0)**
Process	$\nu_\mu e^- \to \mu^- \nu_e$	$\nu_\mu e^- \to \nu_\mu e^-$
Exchange	W^- (s-channel)	Z^0 (t-channel)
$\langle \lvert\mathcal{M}\rvert^2 \rangle$	$16\, G_F^2\, s^2$	$\frac{G_F^2}{16}\left[(c_V{+}c_A)^2 s^2 + (c_V{-}c_A)^2 u^2\right]$
$\frac{d\sigma}{dt}$	$\frac{G_F^2}{\pi}$	$\frac{G_F^2}{256\pi}\left[(c_V{+}c_A)^2 + (c_V{-}c_A)^2 \frac{u^2}{s^2}\right]$
Angular	Isotropic	Forward-peaked
σ_{total}	$\frac{G_F^2\, s}{\pi}$	$\frac{G_F^2\, s}{48\pi}\left[(c_V{+}c_A)^2 + \frac{(c_V{-}c_A)^2}{3}\right]$

Key Takeaways

1. Both cross sections grow linearly with s (total cross section $\sigma \propto G_F^2 s$). This is a feature of the Fermi theory / low-q^2 approximation and signals the breakdown of the theory at $\sqrt{s} \sim M_W$.

2. The charged-current cross section is **isotropic** because only one helicity combination ($LL \to LL$) contributes in the $V{-}A$ theory.

3. The neutral-current cross section has **angular dependence** because $c_V \neq c_A$ for the electron, so both LL and LR helicity combinations contribute.

4. The ratio $\sigma(\nu_\mu e^- \to \nu_\mu e^-)/\sigma(\nu_\mu e^- \to \mu^- \nu_e)$ depends on c_V and c_A, and hence on $\sin^2\theta_W$. This ratio was historically used to measure the Weinberg angle.